

Chapter 1

DIVISION GEOMETRY AND SIZE-DEPENDENT GROWTH REGULATION SHAPE DEVELOPING DROSOPHILA NEUROBLAST LINEAGES IN A CALIBRATED CELLULAR POTTS MODEL

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Author's note

We are currently in the process of preparing this work for submission.

Abstract

Here is the abstract

1.1 Introduction

Here is the introduction.

mud Mutant neuroblasts exhibit a counterintuitive phenotype where they have more neuroblasts but fewer cells (supplemental exp data). The neuroblasts in *mud* mutant lineages originate from symmetric divisions, which yield two neuroblasts of equal size side by side (Figure 1.1 A lower).

Previous models have used a series of ordinary differential equation models to interrogate the system, but those models lack cell-scale variability.

1.2 Results

1.2.1 Spatial organization of the WT neuroblast lineage emerges from calibrated growth and division dynamics.

To establish a baseline WT model for later perturbation analyses, we first calibrated a spatiotemporal CPM-based ABM of the developing *Drosophila* neuroblast lineage. In the WT rule set, the neuroblast divides after growing to $1.166\times$ its initial size. Each neuroblast has an apical axis (Figure 1.1B dashed line) and a default plane of division which passes through a point located 88% down the apical axis from the apical end of the cell and centered laterally (Figure 1.1B top dashed line). By default, that division plane is perpendicular to the apical axis (Figure 1.1B top dashed line), and division-to-division angular variation is sampled from a distribution centered at 0° with a standard deviation of 26° , matching the scale of the experimental WT angle distribution (see supplemental figure). Following division, the larger apical daughter remains a neuroblast, whereas the smaller basal daughter becomes a ganglion mother cell (GMC). The GMC then undergoes one final symmetric division along the cardinal axis that yields the longest new cell membrane (Figure 1.1B bottom) to produce two neurons. Neurons neither grow nor divide. Full parameter values and the calibration workflow are described in Methods.

We anchored this calibration to WT endpoint summaries derived from the experimental dataset in `data/exp/processed/exp_summary.csv`. See Methods for experimental data preprocessing details. In that dataset, WT lineages contained a mean of 44.93 ± 21.68 Pros⁺ cells and exactly one Dpn⁺ neuroblast at the terminal time point. WT lineage area averaged $501.2 \pm 125.8 \mu\text{m}^2$, and because WT lineages end with one neuroblast, total and mean neuroblast area were both $51.9 \pm 42.6 \mu\text{m}^2$. We calibrated the baseline WT condition shown here (`wt_divMean0Stdev26`, `sim61`, `VCV=1`) to match these five metric summaries. Across simulated WT runs, the model produces lineages with a mean area of $525.4 \mu\text{m}^2$, 46.42 Pros⁺ cells, one Dpn⁺ neuroblast, and a mean neuroblast area of $64.2 \mu\text{m}^2$, placing the simulated distribution within one standard deviation of the experimental WT mean for all calibration targets (Figure 1.1D).

Representative simulated WT lineages also qualitatively resemble experimental WT lin-

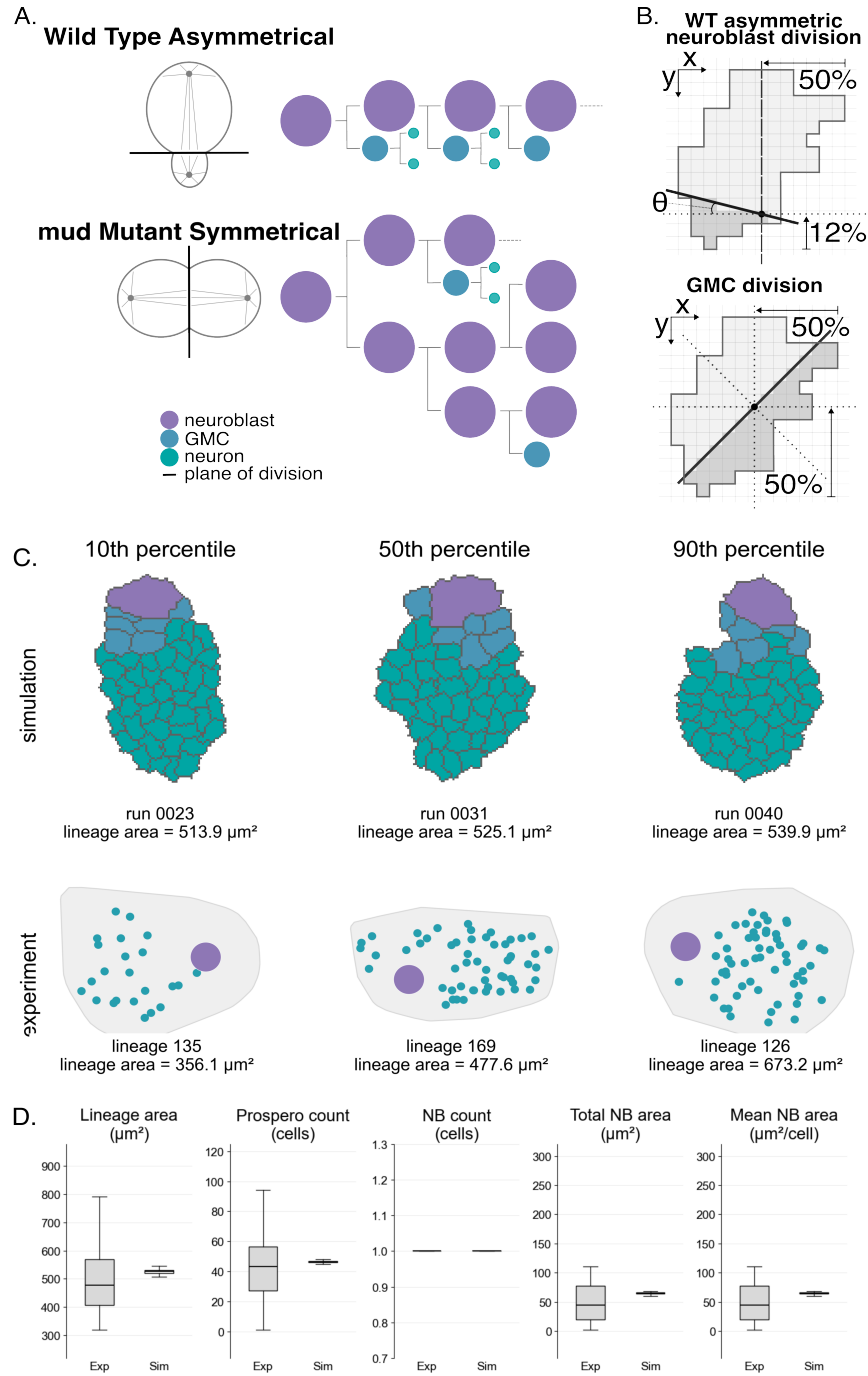


Figure 1.1: Calibrated CPM of developing neuroblast lineage recapitulates expected WT lineage scale, composition, and organization. A. WT and mud Mutant planes of division and differentiation cascades. B. Schematic of how the CPM implements asymmetric division and GMC division. C. Top: Three example simulated WT neuroblast lineages reflecting the 10th, 50th and 90th percentile of lineage area across all WT simulations. Bottom: The distribution of cells in three experimentally imaged WT lineages. Dots represent nuclear centroids. D. Distributions of five calibration metrics across experimental

eages across the observed range of lineage sizes (Figure 1.1C). In both simulation and experiment, the neuroblast typically remains peripheral to the lineage while differentiated GMCs and neurons occupy most of the interior. Importantly, this peripheral organization was not itself a calibration target. Instead, the calibration is performed only against endpoint lineage scale and composition, allowing the spatial organization used in later sections to function as an emergent model behavior rather than a quantity fit by construction.

Together, these results indicate that the calibrated WT CPM captures the expected scale and composition of the lineage while remaining sufficiently unconstrained spatially to serve as a mechanistic testbed for hypotheses about division geometry and lineage organization.

1.2.2 Distinct components of WT division geometry differentially affect lineage organization.

Because spatial organization emerged in the calibrated WT model without being used as a fitting target, we next used that baseline to test which aspects of the imposed division geometry were most important for maintaining the sorted WT lineage morphology. In the model, WT division dynamics contain three separable geometric components. These are the orientation of the apical axis, the amount of rotation applied to the default plane of division, and the degree to which the division plane is offset away from the apical membrane. We systematically perturbed each component while holding the others constant in order to assess its impact on lineage organization. To quantify those effects, we used two spatial readouts. First, we measured normalized heterotypic contact fraction, which reports the extent to which neuroblast and non-neuroblast cell territories intermingle relative to the amount of mixing expected from the lineage composition alone (see Methods). Second, we measured the fraction of the neuroblast perimeter exposed to extracellular space, which reports how peripherally the neuroblast remains positioned within the lineage. Together, these metrics distinguish lineages in which the neuroblast remains spatially segregated at the edge from lineages in which it becomes internalized and more intermixed with differentiated progeny.

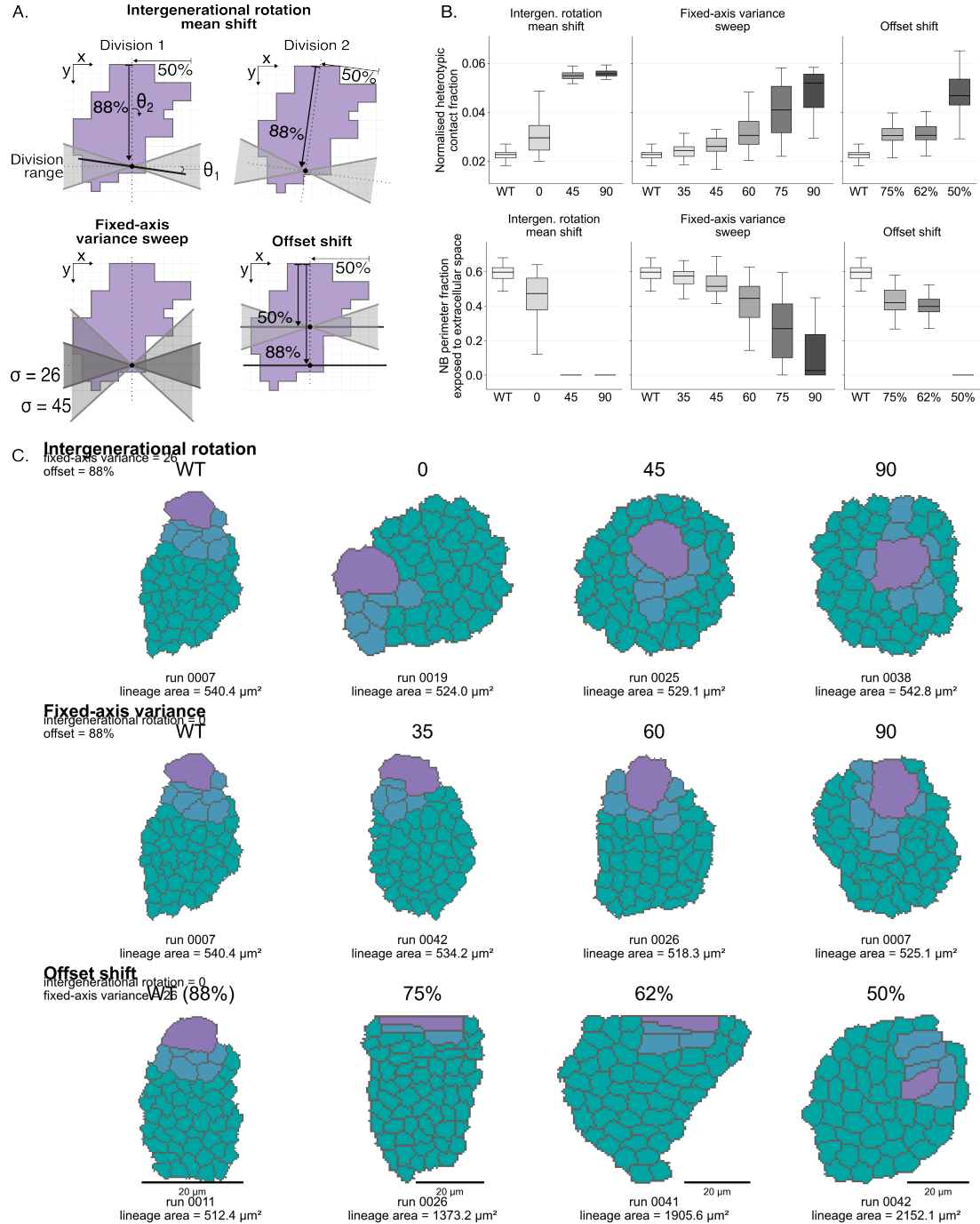


Figure 1.2: Impact of decoupled components of WT division geometry on lineage organization. A. Depictions of intergenerational rotation mean shift (top), fixed-axis variance sweep (bottom left), and offset shift (bottom right) mechanisms implemented to decouple apical axis orientation, stochastic rotation, and offset shift in the CPM. B. Distribution of normalised heterotypic contact values (top) and NB perimeter fraction values (bottom) across different simulated conditions. C. Example lineages drawn from each of the simulated conditions.

Persistent intergenerational rotation disrupts WT lineage sorting.

To test the contribution of persistent intergenerational reorientation, we first varied the amount by which the apical axis rotated from one division to the next (1.2 A top). In these simulations, fixed-axis variance is held at the biologically grounded WT value of 26° , and the division offset is held at the calibrated WT value of 88%. The baseline WT condition uses no persistent mean-shift in intergenerational orientation, whereas perturbed conditions impose apical-axis reorientation drawn from distributions centered at 45° or 90° .

Introducing persistent intergenerational rotation causes a marked loss of the sorted WT morphology (1.2 B left, C top). Relative to the baseline WT condition, lineages with imposed reorientation show substantially higher normalized heterotypic contact and reduced neuroblast exposure to extracellular space, consistent with neuroblast internalization and increased mixing among cell types. This effect is clear at 45° and remains strong at 90° . Thus, within the model, maintaining a stable relative orientation across generations appears to be an important requirement for preserving the peripheral neuroblast position and the spatially sorted architecture of the developing WT lineage.

Increasing fixed-axis variance has limited effects until extreme values.

We next asked whether increasing stochastic variation around a stable WT division axis was sufficient to disrupt lineage organization. To isolate this effect, we returned the intergenerational rotation mean to the baseline WT value of 0° and held the division offset at 88%. We then increased the standard deviation of fixed-axis variation from the biologically grounded WT value of 26° through progressively broader distributions (1.2 A bottom left). This perturbation differs conceptually from the persistent intergenerational rotation tested above. Here, successive divisions continue to fluctuate around the same underlying orientation, rather than drifting persistently away from it over generations.

Across this variance sweep, moderate increases in fixed-axis variability produce only modest changes in lineage organization (1.2 B center, C center). Normalized heterotypic contact rises gradually as the variance increases, but the effect remains substantially smaller than the disruption caused by persistent intergenerational rotation. Likewise, neuroblast

exposure to extracellular space declines progressively across the sweep rather than collapsing immediately. Qualitatively, these lineages largely retain their sorted morphologies, with the neuroblast still positioned near the lineage periphery. Thus, the calibrated WT lineage tolerated appreciable stochastic variation around a stable axis while largely preserving its sorted morphology.

At the highest tested variance (90°), however, this tolerance begins to break down. Under this extreme condition, large rotations of the default division axis produce more mixing and reduced neuroblast exposure. Even so, the resulting disorganization remains weaker than that produced by persistent intergenerational rotation (1.2 B center, C center). These results indicate that stochastic division-angle variability can disrupt lineage organization, but only when it becomes very large. Within the model, these results suggest that preserving a consistent central orientation across generations is more important for maintaining WT spatial organization than tightly minimizing every division-to-division fluctuation around that orientation.

Reducing asymmetric offset produces only modest disorganization relative to intergenerational rotation.

We next asked how strongly lineage organization depends on maintaining the WT asymmetric division offset. To isolate this effect, we held the intergenerational rotation mean at the baseline WT value of 0° and the fixed-axis variance at 26° while shifting the division plane progressively away from the calibrated WT offset of 88% toward more symmetric placements (1.2 A bottom right). We tested offsets of 75%, 62%, and 50%.

Moving the division plane toward the center increases both normalized heterotypic contact and neuroblast internalization, indicating a measurable loss of lineage sorting (1.2 B right, C bottom). However, across this sweep, the magnitude of the effect remains smaller on average than that produced by persistent intergenerational rotation (1.2 B right). Qualitatively, the offset-perturbed lineages also remain fairly organized, with the neuroblast still generally occupying a peripheral position even when it is fully surrounded by cells (1.2 C bottom). Thus we conclude that although maintaining the WT offset contributes to robust

sorting, the model does not suggest that offset alone is the dominant determinant of lineage organization.

One complication in interpreting this perturbation is that reducing the offset increases daughter birth sizes and therefore lineage size. In the smaller simulation domains used initially, these larger lineages frequently encountered the domain boundaries. We therefore repeated the offset-shift simulations in expanded fields of view, although even in those larger domains some lineages still contacted the walls at later times. For that reason, we interpret the offset-shift results more cautiously than the intergenerational-rotation and fixed-axis-variance results. Even with that caveat, both the quantitative spatial readouts and the representative lineages support the same qualitative conclusion: reducing asymmetric offset perturbs sorting, but less strongly than persistent intergenerational rotation.

*1.2.3 Cell-size effects are insufficient to constrain *mud* mutant lineage overgrowth.*

We next used our model to explore which regulatory mechanisms would be sufficient to constrain lineage overgrowth in *mud* mutants. We implemented a *mud* mutant cell type with the same baseline parameters as our calibrated WT simulations, then added a rule for symmetric division: whenever the division plane rotates past 45° from the apical/basal axis, the cell deterministically divides symmetrically along that axis, yielding two neuroblast daughters. Because division-plane rotation is drawn from a normal distribution with mean 0° and standard deviation 26° , this rule causes *mud* mutant neuroblasts to divide symmetrically about 13% of the time. Both this rate and the rotation-based trigger are consistent with experimental observations. We also set the adhesion parameter so that neuroblasts would have stronger adhesion to each other than to all other cells to reflect the experimental observation that neuroblasts in *mud* mutant lineages tend to be next to each other.

We first asked whether the cell-size difference between symmetrically and asymmetrically derived NBs was itself sufficient to constrain mutant lineage overgrowth. Because symmetric division produces two undersized daughters, fixing the division-volume threshold at the value expected for WT NBs would force mutant NBs born of symmetric divisions to spend additional time growing before dividing again, slowing lineage expansion. We therefore

implemented a constant division threshold calibrated to the expected WT division volume (see Methods) and simulated *mud* mutant lineages with all other parameters held at their WT values.

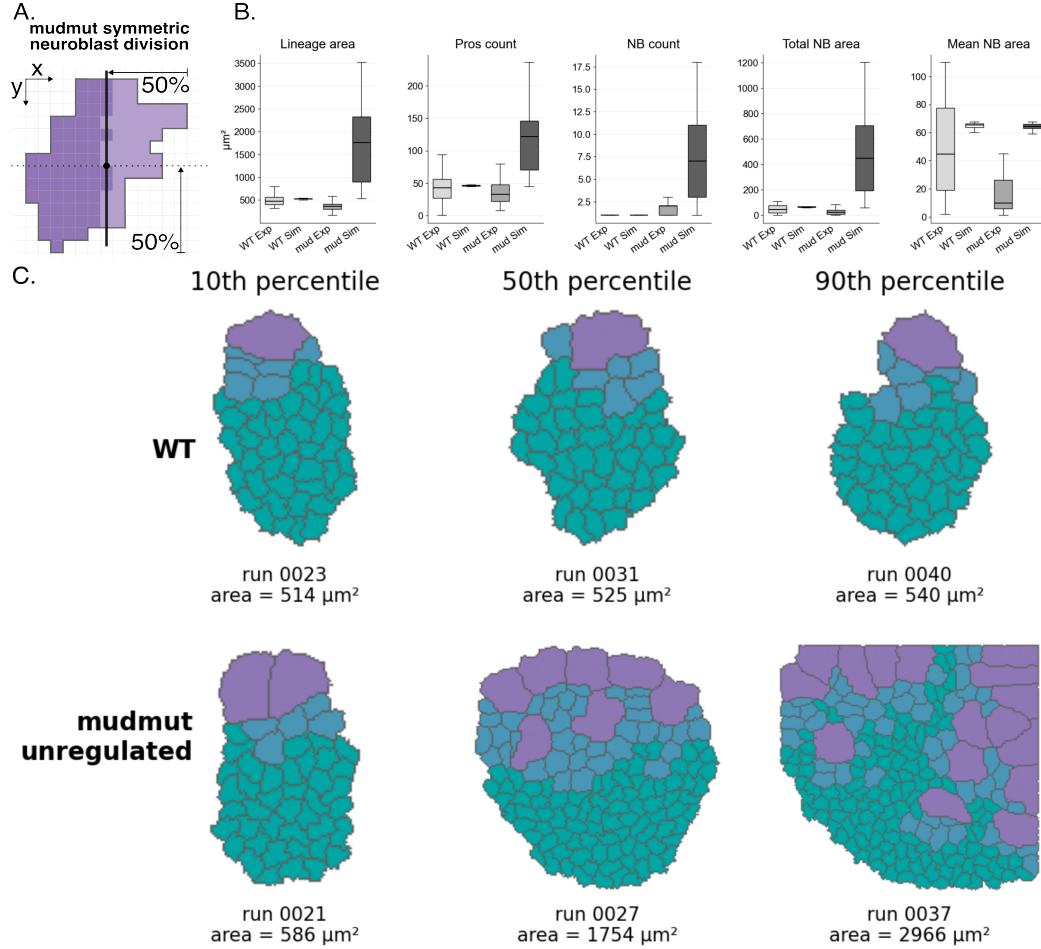


Figure 1.3: **The model predicts that unregulated *mud* mutant lineages should overgrow WT lineages.** **A.** Schematic of an NB undergoing symmetric division as implemented in the model. **B.** Distributions of five non-spatial calibration metrics across WT experimental, WT simulated, *mud* mutant experimental, and unregulated *mud* simulated lineages. **C.** Representative WT and unregulated *mud* mutant simulated lineages at the 10th, 50th, and 90th percentiles of lineage area.

The simulated *mud* mutant lineages outgrew the simulated and experimental WT lineages by all five of our calibration metrics.

1.2.4 Candidate regulatory mechanisms constrain mud mutant overgrowth only when division thresholds scale with birth size.

Two salient differences between *mud* mutant and WT neuroblast lineages are that *mud* mutant lineages have more NBs and that the NBs in *mud* mutant lineages are on average smaller than WT neuroblasts (supplemental data). We were curious whether regulation based on either of those features could constrain mutant lineage overgrowth. To explore this, we implemented four regulatory regimes. In neuroblast volume based (VOL) regulation, we set a constant reference NB volume which is equal to the expected average volume of a WT NB. NB growth rate is repressed if its volume is less than that reference and accelerated if its volume is greater than that reference point. This is exploring the hypothetical scenerio in which growth rate is linked to neuroblast volume and asking whether, in this system, that regime would be sufficient to constrain mutant neuroblast lineage growth. We implemented this regulation in both an ABM-like and PDE-like manner. In the ABM-like implementation, each NB's growth rate is independently regulated by its volume. In the PDE-like implementation, all NBs in the simulation have the same growth rate, and each NB's growth rate is regulated by the average volume across all NBs in the simulation. The PDE-like implementation allows us to explore a hypothetical situation in which the growth rate of NBs is influenced by both the NB's own volume and the volume of the other NBs in the simulation. We also implemented a NB-contact (NB) based regulation. In this regime, NB growth rate is suppressed via a hill function based on how many NB neighbors the NB has. We similarly implemented this in both an ABM and PDE-like manner. In the ABM-like implementation each NB's growth is repressed by its number of NB neighbors and in the PDE-like manner each NB's growth is repressed by the total number of NBs in the simulation. This could represent some kind of diffusible signaling factor.

We applied both of these regulatory regimes with both the constant division threshold (VCV=0) and a dynamic division threshold (VCV=1) in which the NB division volume

threshold is a function of the starting NB volume Figure 1.4 A.

We first confirmed that each of the four candidate regulatory dynamics, when applied to our calibrated WT cell type, leaves the five non-spatial calibration metrics broadly within the WT experimental range (supplement).

When simulated with the fixed division threshold ($VCV=0$), none of the four regulatory mechanisms, simulated *mud* mutant lineages remained well above the experimental *mud* range on lineage area, Pros count, NB count, and total NB area (Fig. 1.4B, light boxes). NB-contact regulation came closest of the four but did not bring all five calibration metrics into agreement with the experimental data.

We then re-ran the same four candidate regulatory dynamics with $VCV=1$, the threshold rule under which each NB divides at a multiple of its own birth size rather than at a shared absolute volume. The biological intuition is that, under this rule, the cell-size memory established by a symmetric division propagates forward across generations: a smaller-born mutant NB has a smaller division threshold, so it divides at a smaller volume, producing daughters that are themselves smaller still (Fig. 1.4A). This keeps *mud* mutant NBs that originate from a symmetric division persistently deeper in the repressive regime imposed by the regulatory dynamic, rather than allowing them to recover to WT-equivalent volumes after a single cycle.

Under this threshold rule, the volume-based regulatory dynamics brought simulated *mud* mutant lineages into the experimental range on lineage area, Pros count, NB count, and total NB area (Fig. 1.4B, dark boxes; Fig. 1.4C,D). Mean NB area is the one metric on which volume-PDE drifts below the experimental range: the smaller per-cycle thresholds produce smaller individual NBs, undershooting the experimental average. NB-contact regulation under $VCV=1$ also tracks the experimental range more closely than it did under $VCV=0$, but remains less precise than the volume-based dynamics across the calibration metrics. The missing ingredient in the unregulated and $VCV=0$ -regulated cases was therefore not a different regulatory mechanism, but a different threshold rule.

Together, these results indicate that constraining *mud* mutant lineage overgrowth in our model requires both a regulatory dynamic that responds to local cellular context and a threshold rule that allows the smaller birth size of symmetric-division descendants to

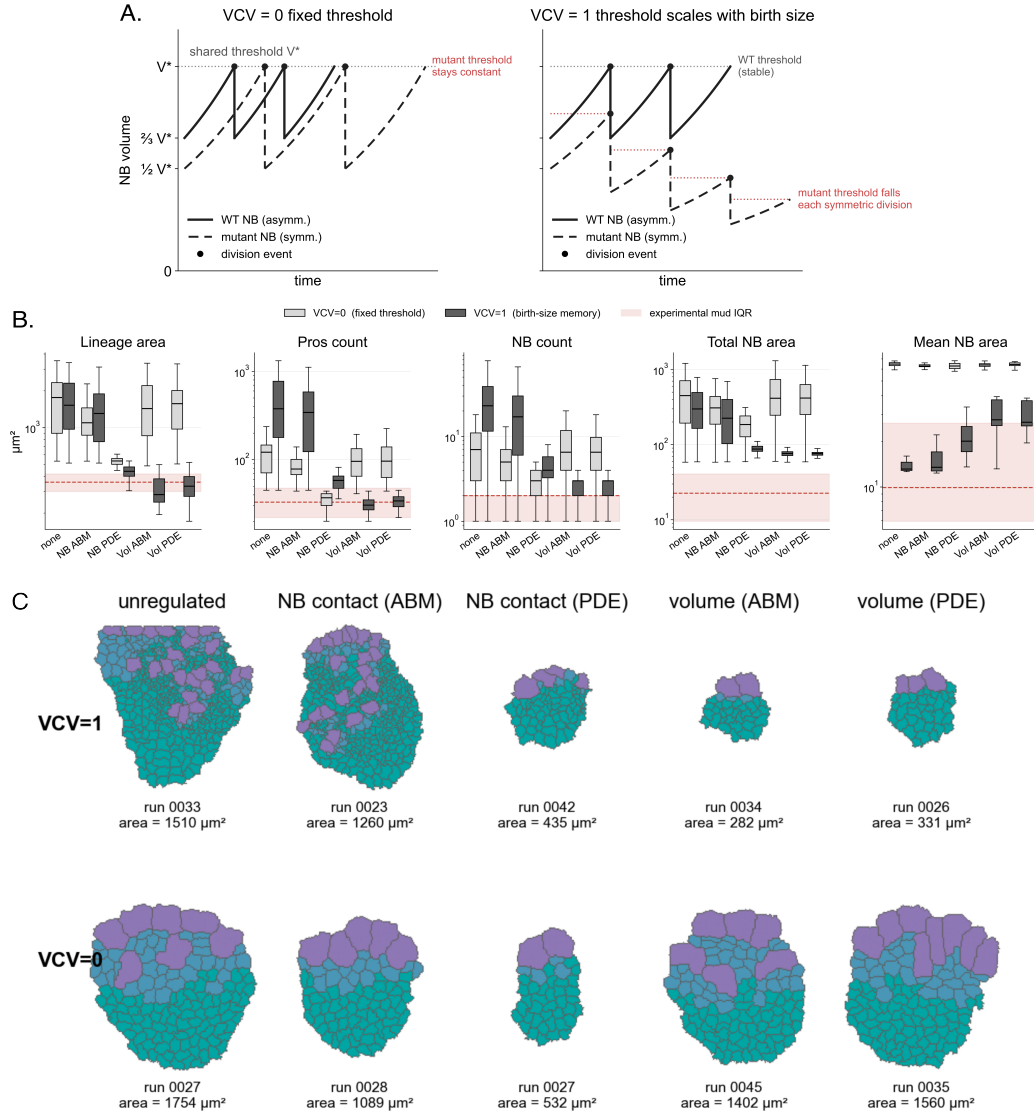


Figure 1.4: Volume-based regulation paired with birth-size-dependent division thresholds constrains *mud* mutant lineage overgrowth. **A.** Schematic contrasting the two threshold rules. **B.** Endpoint distributions of the five calibration metrics across the unregulated baseline plus four candidate regulatory dynamics (NB-ABM, NB-PDE, volume-ABM, volume-PDE), under each threshold rule (light: VCV=0; dark: VCV=1). The red band marks the experimental *mud* interquartile range; the dashed red line marks the experimental median. Y-axes are log-scaled. **C.** Representative simulated *mud* mutant lineages at the 50th percentile of lineage area, organized by regulatory dynamic (columns) and threshold rule (rows).

propagate forward in time.

*1.2.5 Preferential NB–NB adhesion is not required for NB clustering in regulated *mud* mutant lineages*

In the simulated *mud* mutant lineages from Result 4, neuroblasts frequently appeared adjacent to one another, particularly under the volume-based regulatory dynamics that brought lineages closest to the experimental *mud* range. We had initialized these simulations with preferential NB–NB adhesion so we asked whether this adhesion term was responsible for the observed clustering.

We re-ran the volume-based simulations under VCV=1, this time setting the NB–NB adhesion strength equal to all other cell-cell adhesions in the simulation, and compared each run to its with-adhesion counterpart at the same seed.

Removing preferential NB–NB adhesion produced nearly indistinguishable distributions on two of the three spatial metrics: the normalized heterotypic contact fraction, a measure of mixing between NBs and non-NB progeny (Fig. 1.5B), and the homotypic NB contact fraction, the fraction of NB inter-cell contacts that are NB-to-NB (Fig. 1.5D). Both volume-based regulatory dynamics retained roughly 20% of NB inter-cell contacts as NB–NB regardless of adhesion state, indicating that NB clustering in this model emerges from birth geometry and limited subsequent displacement rather than from a preferential adhesion term. Representative paired lineages confirmed this visually (Fig. 1.5A): the spatial layout of NBs and progeny looks essentially the same with or without preferential adhesion.

Together, these results indicate that the NB proximity observed in our model is largely a consequence of birth geometry and limited divisions in constrained mutant lineages and does not require an adhesion-driven mechanism to be maintained.

1.3 Methods

1.3.1 Experimental lineage dataset and preprocessing

Experimental lineage measurements were derived from three-dimensional microscopy segmentations of *Drosophila* larval brain lobes. For each lobe, three VRML (.wrl) files were

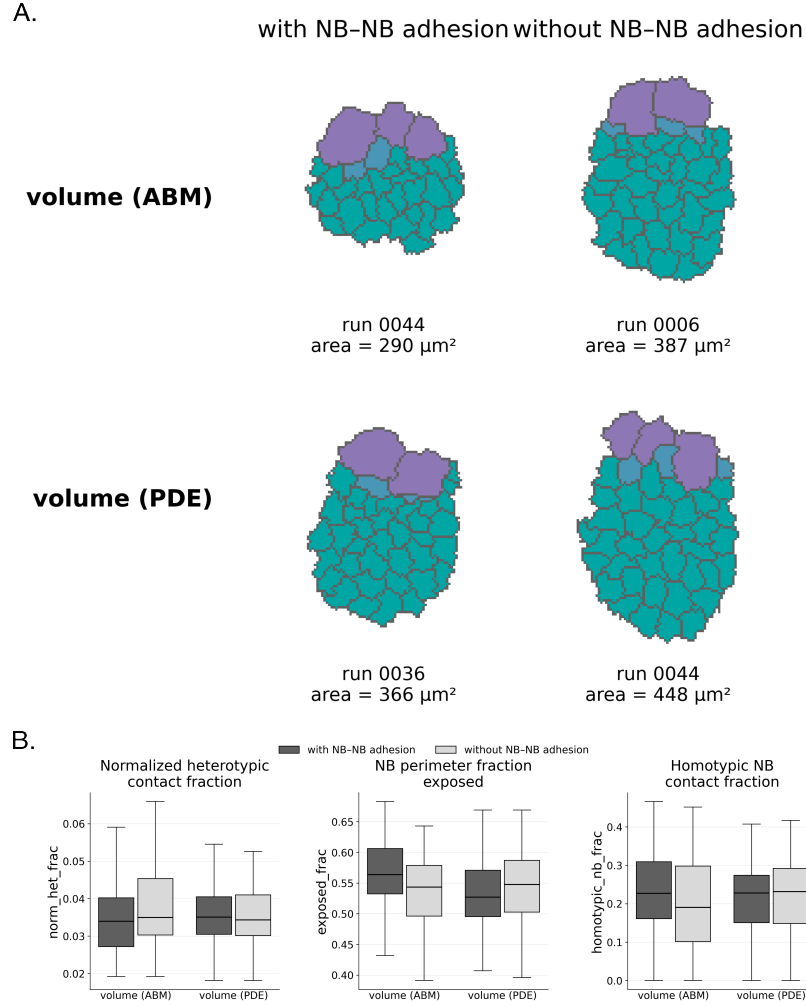


Figure 1.5: **Removing preferential NB-NB adhesion has minimal effect on NB clustering or mixing in regulated *mud* mutant lineages.** **A.** Representative simulated *mud* mutant lineages from each (regulatory dynamic $\times$ adhesion state) combination, picked closest to the mean homotypic NB contact fraction within their group. Rows: regulatory dynamic (volume-ABM, volume-PDE under VCV=1). Columns: with vs without preferential NB-NB adhesion. **B.** Distributions of the normalized heterotypic contact fraction (a measure of mixing between NBs and non-NB progeny) for each regulatory dynamic, the fraction of NB perimeter exposed to extracellular space, and the fraction of NB inter-cell contacts that are NB-to-NB, with and without preferential NB-NB adhesion.

available: lineage hull segmentations, Dpn-positive nuclei, and Pros-positive nuclei. The analyzed dataset included WT control lobes (`lobe1`, `lobe2`, `lobe7`, `lobe8`, `lobe13`, and `lobe15`), *mud* mutant lobes (`lobe110`, `lobe113`, `lobe116`, and `lobe119`), and nanobody perturbation lobes (`lobe16`, `lobe17`, `lobe19`, `lobe20`, `lobe21`, and `lobe22`).

Experimental preprocessing uses the local pipeline implemented in `LINK TO REPO WITH scripts/preprocess_exp.py` and `src/npa/exp-preprocessing/`. WRL files are parsed into triangle meshes by extracting vertex and face blocks from the VRML text and fan-triangulating polygonal faces. Pros-positive cells are assigned to lineage hulls by bounding-box overlap, with assignment requiring at least 95% of vertices to fall within a lineage bounding box. Dpn-positive cells are assigned by sampled mesh containment, using a bounding-box prefilter followed by containment testing of 20 sampled vertices and requiring at least 60% of sampled vertices to lie within a lineage hull.

Lineages are excluded before analysis if they fail either of two quality-control criteria. First, lineages are rejected as disconnected if they contain a secondary connected component with at least 50 faces and volume at least 5.0 in the native WRL coordinate system. Second, lineages are rejected if no Dpn-positive cell can be assigned to the lineage. Rejected lineages are preserved in a separate index but are excluded from all summary analyses. Among WT lineages, 60 of 143 total segmented lineages were retained after filtering (2 lineages were rejected as disconnected and 81 rejected for lacking an assigned Dpn-positive cell). Among *mud* mutant lineages, 59 of 63 total segmented lineages were retained after filtering, and the retained lineages included between 1 and 5 Dpn-positive cells.

For each retained lineage, the lineage hull vertices are projected into two dimensions using the first two principal components of the three-dimensional hull geometry. The projected lineage outline is represented by the convex hull of the projected vertices, buffered by $0.5 \times ds$ with $ds = 0.3 \mu\text{m}$. To generate analysis tensors aligned with the simulation outputs, each projected lineage is rasterized onto a 200×200 canvas. Pixels inside the lineage hull are assigned to the nearest Dpn or Pros centroid using a Voronoi tessellation, yielding a two-channel tensor in which channel 0 represents Dpn territory and channel 1 represents Pros territory. Per-lineage mesh files and stacked genotype-level analysis tensors are written to `data/exp/processed/`.

Experimental endpoint summaries are then computed from the processed lineage dataset. For each lineage we recorded the number of Dpn-positive cells (`n_dpn`), the number of Pros-positive cells (`n_pros`), lineage area in occupied pixels (`lin_area_vox`), total Dpn area (`dpn_area_vox`), and mean Dpn area per lineage (`avg_dpn_area_vox`). Genotype-level means and standard deviations are written to `data/exp/processed/exp_summary.csv` and used as the experimental reference for model calibration and comparison.

1.3.2 Cellular Potts model of WT and mud mutant lineages

All code for this model is publicly available at this url: [ARCADE-URL](#). All code for the analysis run in this manuscript is available at this url: [REPO-URL](#).

We model neuroblast lineage development with a two-dimensional spatiotemporal Cellular Potts model implemented in the ABM framework ARCADE. Although the simulations are visualized and analyzed in two dimensions, ARCADE uses a three-dimensional volume formalism, so growth and division thresholds are specified in μm^3 and represented on a lattice of height one voxel. Unless otherwise noted, simulations run for 2161 ticks with $dt = 0.01667$ h per tick, corresponding to 36 h of total biological time with 1 minute per tick. The default lattice size is $200 \times 200 \times 1$ voxels, with $ds = 0.3 \mu\text{m}$ per voxel edge.

1.3.3 Cellular Potts Hamiltonian

In a Cellular Potts model, each lattice site i is assigned a cell identity σ_i , and each identity belongs to a cell type $\tau(\sigma_i)$. Cell morphology and movement emerge from stochastic updates that consider whether a given lattice site should take its neighbor's identity. Whether a proposed copy is accepted depends on the resulting change in the Hamiltonian energy.

The full Hamiltonian can be written generically as

$$H = H_{\text{adhesion}} + H_{\text{volume}} + H_{\text{surface}}.$$

The adhesion term is

$$H_{\text{adhesion}} = \sum_{\langle i,j \rangle} J(\tau(\sigma_i), \tau(\sigma_j)) (1 - \delta_{\sigma_i, \sigma_j}),$$

where $\langle i, j \rangle$ indexes neighboring lattice sites, σ is the occupying cell identity or extracellular medium, τ is the corresponding cell type, J is the contact-energy coefficient, and $\delta_{\sigma_i, \sigma_j}$ is the Kronecker delta. This term contributes only at interfaces between distinct cells or between a cell and extracellular space.

The volume constraint term is

$$H_{\text{volume}} = \sum_{\sigma} \lambda_v (v(\sigma) - V(\sigma))^2,$$

where $v(\sigma)$ is the current cell volume, $V(\sigma)$ is the target cell volume, and λ_v sets the strength of the volume penalty. The surface constraint term is

$$H_{\text{surface}} = \sum_{\sigma} \lambda_s (s(\sigma) - S(\sigma))^2,$$

where $s(\sigma)$ is the current cell perimeter, $S(\sigma)$ is the target perimeter, and λ_s sets the strength of the surface penalty.

Proposed lattice updates are accepted according to the standard Metropolis rule

$$P(\text{accept}) = \begin{cases} 1, & \Delta H < 0, \\ e^{-\Delta H/T}, & \Delta H \geq 0, \end{cases}$$

where ΔH is the change in Hamiltonian associated with the proposed copy event and T is the simulation temperature.

In the simulations analyzed here, the contact-energy coefficient for cell–extracellular-space interfaces is 80, whereas the default cell–cell coefficient is 50. Because lower contact-energy values correspond to stronger effective adhesion in this formalism, this choice ensured cohesion within each lineage, and prevented cells from dispersing into extracellular space. For mutant simulations that included preferential NB adhesion, the NB–NB contact-energy coefficient was reduced further to 40. Surface lambda values are 0.5 for neuroblasts, 1 for GMCs, and 2 for neurons. Because the contribution of the surface term increases with cell size, these values are chosen so that cells of different types nevertheless maintain comparably smooth and rounded morphologies. The Boltzmann temperature is visually calibrated to 18 so that WT neuroblasts can relax toward rounder shapes between successive asymmetric divisions. Apoptosis is not included.

1.3.4 *WT growth and division rules*

In the WT rule set, the neuroblast grows until it reaches a division threshold equal to $1.166\times$ its initial volume and then divides asymmetrically. The default WT division plane is perpendicular to the apical axis and passes through a point located 88% of the cell length from the apical edge and 50% across the lateral axis. Default angular variation around the default division plane is drawn from a normal distribution centered at 0° with standard deviation 26° . This was based on the evaluation of experimental data that tracked the angle of cell divisions of WT NBs after each successive division (supplemental figure). After WT division, the larger apical daughter remains a neuroblast and the smaller basal daughter becomes a GMC. The GMC divides once more symmetrically to produce two neurons, and neurons neither grow nor divide.

1.3.5 *mud mutant growth and division rules*

To model *mud* mutant lineages, we impose a threshold at $\pm 45^\circ$ such that any sampled rotation of the division plane exceeding 45° in magnitude is converted into a deterministic symmetric division along the neuroblast apical axis. In the mutant rule set used here, this yields approximately 13% symmetric divisions, consistent with experimental observation. Mutant daughter neuroblasts inherit apical axes rotated relative to the parent axis by an amount drawn from a normal distribution with mean 0° and standard deviation 30° . This inheritance rule is chosen to approximate the qualitative loss of stable spindle orientation observed experimentally in *mud* mutant lineages.

Imposed and volume-sensitive division thresholds

The division threshold for *mud* mutants is implemented in two different ways. Under the fixed-threshold ruleset (VCV=0), all neuroblasts of a given type share the same critical volume V_c , and division occurs when the cell reaches

$$V \geq \text{SIZE_TARGET} \times V_c.$$

In practice, this means *mud* mutant cells divide when they surpass the same absolute threshold, which is set to $1.166\times$ the expected WT neuroblast initial volume.

Under the birth-volume-dependent ruleset ($\text{VCV}=1$), daughter critical volume is reset from daughter birth volume. For daughter neuroblasts, the new critical volume is

$$V_{c,\text{daughter}} = \max(V_{\text{birth}}, 0.2V_{c,\text{WT}}),$$

where $V_{c,\text{WT}}$ is the baseline WT neuroblast critical volume. For GMC daughters, the analogous rule is

$$V_{c,\text{GMC}} = \max(V_{\text{birth}}, 0.1V_{\text{NB,birth}}).$$

Under this ruleset, the division threshold decreases as mutant daughter cells are born progressively smaller after successive symmetric divisions, but it cannot fall below a fixed minimum of 20% of the WT neuroblast critical volume (or 10% the initial GMC critical volume for the GMCs). Biologically, this corresponds to the assumption that a cell can only divide if it remains above a minimum size. Thus, $\text{VCV}=0$ tests a fixed-threshold size hypothesis, whereas $\text{VCV}=1$ allows daughter birth size to propagate into the next division threshold.

NB growth regulatory mechanisms

To test candidate mechanisms constraining mutant lineage growth, we vary whether neuroblast growth rate is regulated by cell volume or by NB–NB contact. Each candidate repression mechanism can operate either in a cell-autonomous ABM-like mode or in a population-averaged PDE-like mode.

Volume-regulated growth. When volume-sensitive growth is enabled, the effective growth rate is scaled as a power law of cell volume relative to a reference volume. In the ABM-like implementation, each cell updates its own growth rate according to

$$r_i = r_0 \left(\frac{V_i}{V_{\text{ref}}} \right)^s,$$

where r_0 is the baseline growth rate, V_i is the current cell volume, V_{ref} is the expected equilibrium volume for that cell type, and s is the volume-sensitivity exponent. In all volume-regulated simulations analyzed here, $s = 3$. For neuroblasts, the reference volume

is computed from the midpoint of the WT-like asymmetric division cycle,

$$V_{\text{ref,NB}} = \frac{V_{\text{div}}(1 + f_{\text{retain}})}{2},$$

where $V_{\text{div}} = \text{SIZE_TARGET} \times V_{c,\text{WT}}$ and f_{retain} is the fraction of pre-division volume retained by the NB after division. For GMCs, the reference volume is

$$V_{\text{ref,GMC}} = \frac{V_c + \text{SIZE_TARGET} \times V_c}{2} = V_c \frac{1 + \text{SIZE_TARGET}}{2}.$$

In the PDE-like implementation, all cells of a given type share a common growth rate based on the population-average volume rather than their own individual volume. This is implemented by setting `PDELIKE=1`; the ABM-like implementation uses `PDELIKE=0`. For neuroblasts this becomes

$$r_{\text{NB}} = r_0 \left(\frac{\bar{V}_{\text{NB}}}{V_{\text{ref,NB}}} \right)^s,$$

where $\bar{V}_{\text{NB}}$ is the mean neuroblast volume across the simulation. For GMCs the analogous update is

$$r_{\text{GMC}} = r_0 \left(\frac{\bar{V}_{\text{GMC}}}{V_{\text{ref,GMC}}} \right)^s,$$

with $\bar{V}_{\text{ref,GMC}} = \bar{V}_c(1 + \text{SIZE_TARGET})/2$. Biologically, this PDE-like formulation can be interpreted as a population-wide signal that causes all cells of a given type to respond to the mean state of that population.

NB-contact-regulated growth. When NB self-repression is enabled, growth rate is instead downregulated by a Hill-type function of neuroblast contact. In the ABM-like implementation, each neuroblast responds to its own number of unique NB neighbors:

$$r_i = r_0 \frac{K^n}{K^n + N_i^n},$$

where N_i is the number of NB neighbors for cell i , K is the half-maximal neighbor count, and n is the Hill exponent. In all NB-contact-regulated simulations analyzed here, $K = 2$ neighbors and $n = 3$. This gives maximal growth in the absence of neighboring neuroblasts and progressively stronger repression as local NB crowding increases.

In the PDE-like implementation, all neuroblasts share a common growth rate determined by a population-level NB signal. As above, this is implemented by setting `PDELIKE=1`; the

ABM-like NB-contact implementation uses PDELIKE=0. In the implementation used here, that signal is the number of neuroblasts in the simulation rather than each cell’s local neighbor count, yielding

$$r_{\text{NB}} = r_0 \frac{K^n}{K^n + N_{\text{pop}}^n},$$

where N_{pop} is the population-level NB signal. This PDE-like variant therefore acts as a global repression regime in which all neuroblasts slow together as the mutant NB population expands.

1.3.6 Model calibration and simulation subsets

The WT baseline is calibrated against experimental endpoint summaries rather than against spatial organization. Calibration targets are lineage area, number of Pros-positive cells, number of Dpn-positive cells, total Dpn area, and mean Dpn area per lineage at the terminal time point. These quantities are compared to the averages calculated from the WT experimental data. Additional calibration constraints require WT neuroblast cycle times below 90 min and GMC cycle times of at least 8 h, both also experimentally observed metrics.

Offset-perturbed WT simulations require partial recalibration because shifting the division plane toward a more symmetric split changes daughter birth sizes and therefore alters lineage-scale outcomes even when the cell identity rules are unchanged. For these offset-perturbed conditions, NB and GMC growth rates are recalibrated to maintain approximately comparable terminal cell counts across conditions, rather than to preserve the WT volume-based metrics or cell-cycle-duration targets. This choice is made so that differences in lineage mixing can be interpreted more fairly, because the opportunity for mixing depends in part on how many cells a lineage contains. We note that the offset-perturbed simulations also produce larger cells, which in principle could increase the amount of neuroblast coverage and thereby affect interface-based spatial readouts. However, in practice neuroblast coverage remains limited in these simulations, so we interpret the observed changes in mixing primarily as consequences of the altered division geometry rather than of expanded neuroblast area alone. This was done to try to provide a fair qualitative comparison of

mixing between lineages. Because cells move as a unit, the amount of mixing a lineage can have is proportional to the number of cells it contains. We acknowledge that, because the offset-perturbed simulations had cells of larger volume, this provides the opportunity for more neuroblast coverage. However we found that there wasn't that much neuroblast coverage anyways so didn't worry about it.

1.3.7 Quantitative measures for simulation-experiment comparison

All quantitative analyses are performed on terminal snapshots. For both experimental and simulated lineages, occupied pixel area is converted to projected area in μm^2 by multiplying by $ds^2 = 0.09 \mu\text{m}^2$ per pixel whenever areas are reported in physical units.

Endpoint composition and scale metrics

The primary non-spatial comparison metrics are: `lin_area_vox`, the total occupied lineage area in pixels; `n_pros`, the number of Pros-positive progeny; `n_dpn`, the number of Dpn-positive neuroblasts; `dpn_area_vox`, the total Dpn area in the lineage; and `avg_dpn_area_vox`, the mean Dpn area per lineage. For experimental lineages, Dpn and Pros areas are computed from the Voronoi territories of the processed lineage tensors. For simulations, these metrics are computed directly from raw `CELLS.json` and `LOCATIONS.json` voxel lists at the last timepoint. Simulation outputs are summarized either as per-run endpoint values or as condition-level means and standard deviations, depending on the figure.

Parameters

Core parameters for the calibrated WT baseline, the offset-perturbed WT simulations, and the mutant growth-regulation simulations are summarized in Tables 1.1, 1.2, and 1.3. The full parameterization logic, calibration sweeps, and offset-specific recalibration details are documented in `ARCADE/PARAMETERS_EXPLAINED.md` and `ARCADE/PARAMETERS_y50_EXPLAINED.md`.

Table 1.1: Core WT model parameters used in the calibrated baseline simulations.

Parameter	Value	Units	Role
<code>ds</code>	0.3	$\mu\text{m}/\text{voxel}$	Spatial scale of the lattice
<code>dt</code>	0.01667	h/tick	Temporal scale (≈ 1 min/tick)
<code>ticks</code>	2161	ticks	Total simulated duration (≈ 36 h)
<code>length, width, height</code>	200, 200, 1	voxels	Two-dimensional simulation domain
<code>NB CRITICAL_VOLUME</code>	17.5	μm^3	Resting neuroblast target volume
<code>NB SIZE_TARGET</code>	1.166	fold	Neuroblast divides at $1.166\times$ resting size
<code>NB CELL_GROWTH_RATE</code>	2.25	$\mu\text{m}^3/\text{h}$	Sets WT neuroblast cycle length
<code>GMC CELL_GROWTH_RATE</code>	0.34	$\mu\text{m}^3/\text{h}$	Sets GMC cycle length
<code>Neuron CELL_GROWTH_RATE</code>	0	$\mu\text{m}^3/\text{h}$	Neurons do not grow
<code>WT division offset</code>	88	% of apical axis length	Asymmetric NB/GMC split position
<code>WT division-angle variation</code>	$N(0, 26)$	degrees	Stochastic variation around the default division axis
<code>ADHESION</code>	80	a.u.	Generic cell-cell adhesion
<code>TEMPERATURE</code>	18	a.u.	Potts acceptance temperature

Table 1.2: Parameters changed for the WT offset perturbation relative to the calibrated WT baseline. Parameters not listed were unchanged from Table 1.1.

Parameter	Baseline WT	Offset-50 WT	Reason for change
WT division offset	88%	50%	Perturbation under study
NB SIZE_TARGET	1.166	2.0	Maintains return to near-resting size after a 50/50 split
NB CRITICAL_VOLUME	$17.5 \mu\text{m}^3$	$13 \mu\text{m}^3$	Keeps average NB projected area near the WT target
GMC CRITICAL_VOLUME	$11 \mu\text{m}^3$	$13 \mu\text{m}^3$	Matches larger GMC birth size after symmetric splitting
Neuron CRITICAL_VOLUME	$11 \mu\text{m}^3$	$13 \mu\text{m}^3$	Matches inherited neuron birth size
GMC CELL_GROWTH_RATE	$0.34 \mu\text{m}^3/\text{h}$	$1.53 \mu\text{m}^3/\text{h}$	Preserves approximate GMC cycle duration after the offset shift

Table 1.3: Additional parameters used in mutant growth-regulation simulations.

Parameter	Value	Units	Role
VOLUME_BASED_ CRITICAL_VOLUME	0 or 1	—	Selects imposed (VCV=0) or birth-volume-dependent (VCV=1) division thresholds
PDELIKE	0 or 1	—	Selects ABM-like (0) or population-averaged PDE-like (1) regulation
DYNAMIC_GROWTH_RATE_ VOLUME	0 or 1	—	Enables or disables volume-regulated growth
GROWTH_RATE_VOLUME_ SENSITIVITY	3.0	—	Exponent s controlling the strength of volume-dependent growth repression
DYNAMIC_GROWTH_RATE_NB_ SELF_REPRESSION	0 or 1	—	Enables or disables NB-contact-regulated growth
NB_CONTACT_ HALF_MAX	2.0	neighbors	Half-maximal repression constant K for NB-contact regulation
NB_CONTACT_HILL_N	3.0	—	Hill exponent n controlling NB-contact repression steepness

Normalised heterotypic contact fraction

To quantify lineage mixing, we compute a normalised heterotypic contact fraction from each simulated lineage snapshot. This choice is motivated by the Cellular Potts formalism itself, in which heterotypic interfaces are directly related to the contact-energy term in the Hamiltonian and mismatched cell contacts are a standard readout of cell sorting behavior (DOI: 10.1103/PhysRevE.47.2128; DOI: 10.1371/journal.pcbi.1008576). Pixels are classified as neuroblast (NB), non-NB progeny, or empty. All horizontally and vertically adjacent occupied pixel pairs are then enumerated using 4-connectivity. A contact is considered heterotypic if one pixel is NB and the other is non-NB, and homotypic otherwise. The raw heterotypic contact fraction is defined as

$$\text{het_frac} = \frac{\text{heterotypic contacts}}{\text{heterotypic contacts} + \text{homotypic contacts}}.$$

Because this metric is computed from occupied pixel-pixel contacts, we normalize the raw heterotypic contact fraction by the value expected under random pixel-label mixing across occupied lineage territory, while preserving the observed NB and non-NB area fractions. If p_{NB} is the fraction of occupied pixels belonging to the neuroblast and p_{nonNB} is the fraction belonging to non-neuroblast cells, then the expected heterotypic fraction under this voxel-scale random-mixing null is $2p_{\text{NB}}p_{\text{nonNB}}$. The normalized metric is therefore

$$\text{norm_het_frac} = \frac{\text{het_frac}}{2p_{\text{NB}}p_{\text{nonNB}}}.$$

Values less than 1 indicate greater segregation than expected from voxel-scale random label shuffling, values near 1 indicate approximately random interfacial mixing, and values greater than 1 indicate excess intermixing.

Neuroblast exposure to extracellular space

To quantify how peripherally the neuroblast is positioned, we measure the fraction of the neuroblast perimeter exposed to extracellular space. Using the NB mask from each simulated lineage, we enumerate all horizontal and vertical NB boundary contacts. A boundary segment is counted as exposed if it separates NB from empty space and as internal contact

if it separates NB from a non-NB cell. The neuroblast exposure metric is defined as

$$\text{exposed_frac} = \frac{\text{NB-empty boundary contacts}}{\text{NB-empty boundary contacts} + \text{NB-nonNB boundary contacts}}.$$

Higher values indicate a more peripheral neuroblast, whereas lower values indicate increasing neuroblast internalization within the lineage.

Representative lineage selection

For the representative WT lineages shown in Figure 1, simulated runs and experimental lineages are selected by lineage-area percentile. We identify examples closest to the 10th, 50th, and 90th percentiles of the WT lineage-area distributions. For the WT geometry perturbation figures, representative simulated runs are chosen from the corresponding condition-specific endpoint distributions, with manual overrides applied only when necessary to avoid a misleading representative example for a given spatial phenotype.